SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)
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॥वसुधैव कुटुम्बकम्॥

CSE - Academic Year 2026-27
Java Programming Lab | Batch 2025-29

ASSIGNMENT : 22

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Batch	CSE A - 2
Class	Java Programming Lab
Academic Year & Semester	2025-29 : Semester 3
Date of Submission	

TITLE : Demonstration and exploration of JDBC classes and interfaces (PreparedStatement, and ResultSet).

Learning Objectives :

To implement PreparedStatement and ResultSet for secure query execution and data retrieval.

Theory : PreparedStatement is a JDBC interface used to execute precompiled SQL statements with parameters. It uses **?** as placeholders for values, which are supplied using methods such as **setInt()** and **setString()**. It makes SQL queries easier to execute with different values and helps protect applications from SQL injection.

ResultSet is a JDBC interface used to store the records returned by a SELECT query. It provides methods such as **next(), getInt(), getString(), and getDouble()** to access the retrieved data.

PreparedStatement is used to execute parameterized queries using methods such as **executeQuery()** and **executeUpdate()**, while **ResultSet** is used to read the records returned by SELECT queries. Together, they provide a secure and efficient way to interact with a database using JDBC.

Exercise :

1. Create a login application using PreparedStatement to check username and password.

```
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import java.sql.ResultSet;
import java.util.Scanner;

public class java21a {
    private static final String DB_URL =
"jdbc:mysql://localhost:3306/hospital_db";
    private static final String DB_USER = "root";
    private static final String DB_PASSWORD = "anushree";

    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);

        System.out.println("=== System Login ===");
        System.out.print("Enter Username: ");
        String username = scanner.nextLine();

        System.out.print("Enter Password: ");
        String password = scanner.nextLine();

        // SQL Query with placeholders (?) to prevent SQL Injection
        String query = "SELECT * FROM users WHERE username = ? AND
password = ?";

        try (Connection conn = DriverManager.getConnection(DB_URL,
DB_USER, DB_PASSWORD);
             PreparedStatement pstmt = conn.prepareStatement(query)) {

            // Set parameters
            pstmt.setString(1, username);
            pstmt.setString(2, password);

            try (ResultSet rs = pstmt.executeQuery()) {
```

```
                if (rs.next()) {
                    System.out.println("\nLogin Successful! Welcome, " +
rs.getString("username") + ".");
                } else {
                    System.out.println("\nInvalid Username or Password.
Access Denied.");
                }
            }
        } catch (Exception e) {
            System.err.println("Database Error: " + e.getMessage());
        }
        scanner.close();
    }
}
```

In SQL :

```
CREATE TABLE users (
    id INT PRIMARY KEY AUTO_INCREMENT,
    username VARCHAR(50) NOT NULL,
    password VARCHAR(50) NOT NULL
);
INSERT INTO users (username, password) VALUES ('admin', 'admin123');
INSERT INTO users (username, password) VALUES ('anushree', 'anu123');
```

OUTPUT :

```
=== System Login ===
Enter Username: admin
Enter Password: ad12345

Invalid Username or Password. Access Denied.
```

—

```
=== System Login ===
Enter Username: anushree
Enter Password: anu123

Login Successful! Welcome, anushree.
```

2. Create a Java application using **PreparedStatement** to authenticate **hospital staff login** (doctor/nurse/login ID and password). Display appropriate access messages based on authentication result.

```
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import java.sql.ResultSet;
import java.util.Scanner;

public class HospitalStaffLogin {
    private static final String DB_URL =
"jdbc:mysql://localhost:3306/hospital_db";
    private static final String DB_USER = "root";
    private static final String DB_PASSWORD = "anushree";

    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);

        System.out.println("=== Hospital Staff Portal ===");
        System.out.print("Enter Staff ID: ");
        String staffId = scanner.nextLine();

        System.out.print("Enter Password: ");
        String password = scanner.nextLine();

        // Parameterized query for security and precompilation
        String query = "SELECT role FROM hospital_staff WHERE staff_id =
? AND password = ?";

        try (Connection conn = DriverManager.getConnection(DB_URL,
DB_USER, DB_PASSWORD);
             PreparedStatement pstmt = conn.prepareStatement(query)) {

            pstmt.setString(1, staffId);
            pstmt.setString(2, password);

            try (ResultSet rs = pstmt.executeQuery()) {
                if (rs.next()) {
                    String role = rs.getString("role");
                    System.out.println("\nAuthentication Successful!");
```

```
                    // Display role-specific access messages
                    if ("Doctor".equalsIgnoreCase(role)) {
                        System.out.println("Access Granted: Welcome, Dr. " + staffId + ". Redirecting to Doctor Dashboard (Patient Records & Prescriptions).");
                    } else if ("Nurse".equalsIgnoreCase(role)) {
                        System.out.println("Access Granted: Welcome, Nurse " + staffId + ". Redirecting to Nursing Station (Vitals & Ward Management).");
                    } else {
                        System.out.println("Access Granted: Welcome, Staff member.");
                    }
                } else {
                    System.out.println("\nAuthentication Failed: Invalid Staff ID or Password.");
                }

                System.out.println();
            }

        } catch (Exception e) {
            System.err.println("Database Error: " + e.getMessage());
        }
        scanner.close();
    }
}
```

In SQL :

```
CREATE TABLE hospital_staff (
    id INT PRIMARY KEY AUTO_INCREMENT,
    staff_id VARCHAR(50) UNIQUE NOT NULL,
    password VARCHAR(50) NOT NULL,
    role VARCHAR(20) NOT NULL -- 'Doctor' or 'Nurse'
);
INSERT INTO hospital_staff (staff_id, password, role) VALUES
('DOC101', 'docpass', 'Doctor'),
('NUR201', 'nurpass', 'Nurse');
```

OUTPUT :

```
=== Hospital Staff Portal ===
Enter Staff ID: NUR201
Enter Password: nurpass

Authentication Successful!
Access Granted: Welcome, Nurse NUR201. Redirecting to Nursing Station (Vitals & Ward Management).
```

```
=== Hospital Staff Portal ===
Enter Staff ID: DOC101
Enter Password: docpass

Authentication Successful!
Access Granted: Welcome, Dr. DOC101. Redirecting to Doctor Dashboard (Patient Records & Prescriptions).
```

```
=== Hospital Staff Portal ===
Enter Staff ID: DOC102
Enter Password: doc1234

Authentication Failed: Invalid Staff ID or Password.
```

Outcome :

- Understood the use of **PreparedStatement** and **ResultSet** in JDBC.
- Learned to execute parameterized SQL queries using PreparedStatement.
- Learned to retrieve and process records using ResultSet.
- Understood the use of **setInt(), setString(), executeQuery(), and next()** in JDBC.
- Successfully implemented username/password and hospital staff authentication using JDBC.

Conclusion :

This assignment demonstrated the use of **PreparedStatement** and **ResultSet** for secure database interaction using JDBC. Login and hospital staff authentication were successfully performed using parameterized SQL queries and ResultSet for retrieving matching records. The assignment helped in understanding secure query execution and basic data retrieval in a Java application connected to a database.